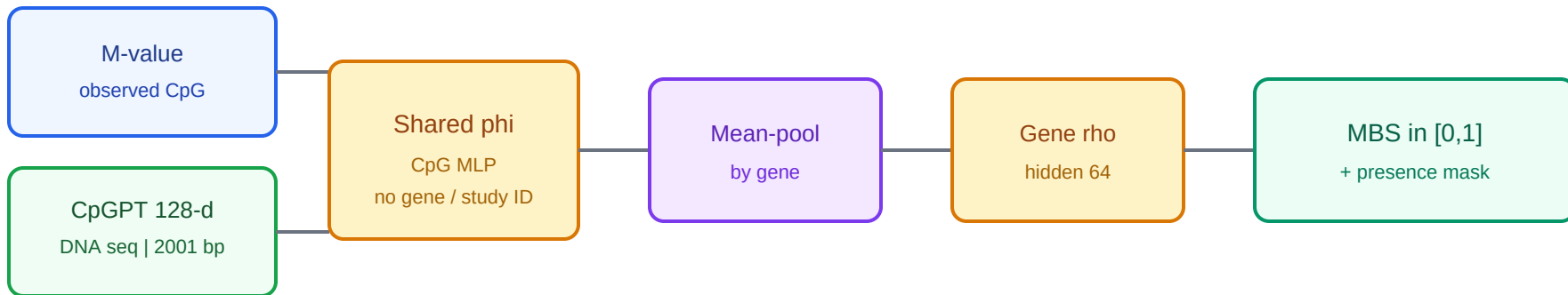


N-light@64 + CpGPT - light product path

FlatDeepSetRegion | gene-linked CpGs | HM450 nine-pack | product readout = frozen nested elastic-net



Product readout (freeze encoder)

Frozen nested elastic-net on sample x gene MBS -> age / tissue / sex

Joint mbs_e2e heads = architecture screen only (noisier; worse age)

Locked metrics (nine-pack HM450)

5x6 OOF nested (65k prefix, 2 646 genes): tissue F1 0.316 | age MAE 9.59 | sex AUROC 0.822

+ CpGPT fold-0 x6: age MAE 8.10 +/- 0.14 (vs 9.57 baseline) | sex +0.065 | tissue -0.021